1 Objectives and Significance

Suppose one is asked, "Are the paintings of Leonardo da Vinci more like those of Peter Paul Rubens or those of Raphael?" One is likely to respond with: "What is the basis for the comparison?" Indeed, without knowing the criteria (i.e., the query concept) by which a comparison is to be made, a database system cannot conduct the search. For relational database searches, query concepts are conveyed via query languages. For Internet document searches, query concepts are conveyed via keywords. For a multimedia search task, however, a query concept is hard to articulate, and articulation can be subjective. For instance, in an image search, it is difficult for a user to describe a desired image using low-level features such as color, shape and texture (these are widely used features for representing images [69]). In addition, different users may perceive the same image differently. Even if an image is perceived similarly, users may use different vocabulary (i.e., different combinations of low-level features) to depict it. Furthermore, most users are not trained to specify simple query criteria using, for example, Boolean algebra. In order to make information access easier and more personalized, it is both necessary (for capturing subjective concepts) and desirable (for alleviating frustrations of users in specifying complex query concepts) to build intelligent search engines that can quickly *learn* users' query concepts through *relevance feedback*.

Traditional learning and relevance feedback techniques, unfortunately, are not suitable for online query-concept learning for at least two reasons.

- *Time and sample constraints*. Traditional learning methods such as decision trees and neural networks require a large number of training instances (i.e., samples) and can take a long time (more than a few seconds) to learn a concept [7, 46, 57]. But, online users are typically impatient and cannot be expected to wait around or to provide a great deal of feedback.
- *Seeding constraint*. All traditional relevance feedback methods [43, 64, 86] require users to provide "good" examples to seed a query. However, finding good seeds is the job of the search engine itself, and this circular requirement leaves the core problem— learning users' query concepts—unsolved.

The goal of the proposed research plan is to make fundamental advances towards intelligent search engines through the development of online query-concept learners. The specific targets are as follows:

1. To design *novel learning algorithms* that grasp a user's query concept quickly despite time, sample, and seeding constraints.
2. To develop techniques that can detect *concept drift* during a relevance feedback session, and to handle concept drift in the learning algorithms. (See task C in Section 3.2.)
3. To devise *multi-resolution image characterization methods* for improving both search accuracy and search efficiency. (See task D in Section 3.2.)
4. To ensure the *scalability in feature dimension, dataset size, and concept complexity* of the developed learning algorithms. (See tasks E and F in Section 3.2.)
5. To conduct *validation* on developed learning algorithms with experimental data provided by colleagues at IBM Almaden Research Center, IBM T.J. Watson Research Center, and Benchthalon. (See Section 3.3.)

Two learning algorithms we have recently developed, MEGA [17, 49] and SVM_{Active} [77], provide a good foundation for our research team to continue making progress. In addition, the prototype [15, 16] that we

have implemented allows us to quickly examine new hypotheses (some are described in Section 3.2), and to solicit input from the research community and from our partners. To evaluate how fast and how well our developed algorithms can capture subjective query concepts, we have established several image testbeds. We are also participating in efforts [14] for establishing standard tests [1] for evaluating image and video retrieval systems.

The project's broader impacts upon information retrieval are potentially substantial. First, rapid proliferation of multimedia content in digital libraries and on the Web underscores the increasing importance of having effective multimedia search tools. Second, intelligent query-concept learners will directly or indirectly make traditional text-based information retrieval easier and more personalized. Directly, a text collection can employ an intelligent learner to better capture users' query concepts. Indirectly, for instance, multimedia data can be added to a text collection so that searches can be conducted through interfaces that contain pictures and graphics. Even young students who have not learned Boolean algebra can use images and graphics to search for stories and books. In addition to bringing benefits to education, we believe that this research project will contribute to making information more accessible for underprivileged users who are not yet able to enjoy the full benefits of the information revolution.

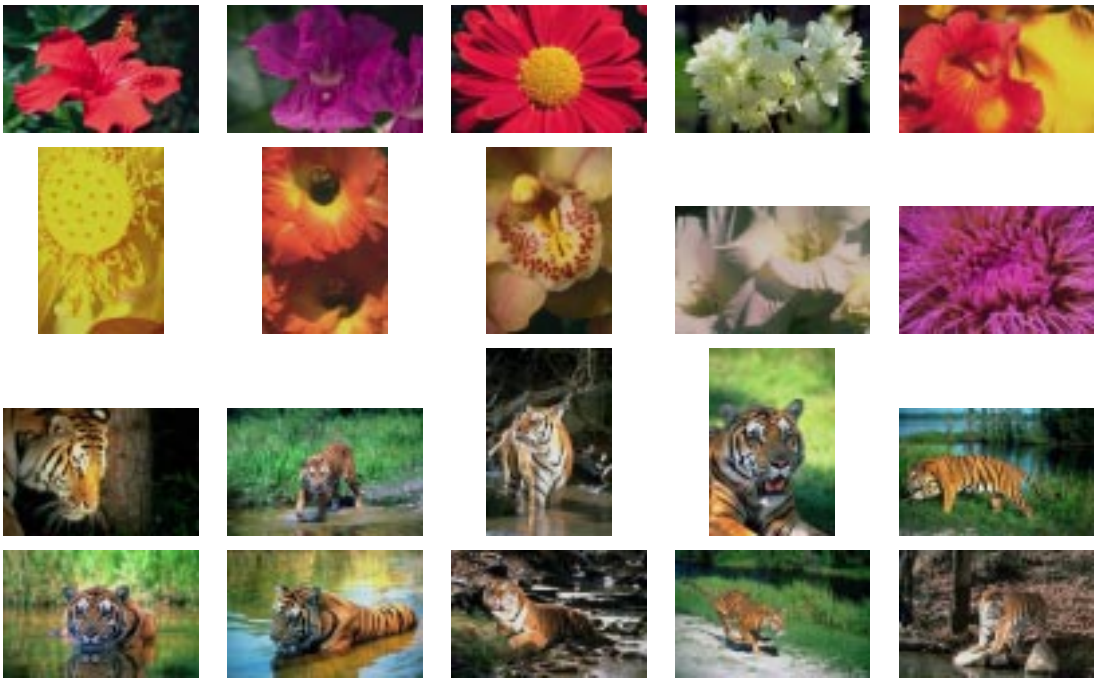


Figure 1: Flowers and Tigers Sample Query Results.

Illustrative Example

This example shows how our prototype [15, 16] that employs MEGA and SVM_{Active} works. Our prototype does not require seeding a query. It presents randomly selected images as the first round of examples. Even if all of the images generated in the first round are marked irrelevant by the user, our learning algorithm (i.e., MEGA) uses the irrelevant images to reduce the set of potentially relevant images substantially. Thus, the probability is higher that a relevant image is sampled and presented to the user for feedback in the next round.

Figure 1 shows two sample results after three rounds of relevance feedback from our prototype: one from a top-10 flowers query, and one from a top-10 tigers query. The returned images vary somewhat in color, shape, and background. The returned flowers have colors of red, purple, white, and yellow, with or without leaves. The returned tiger images have tigers of different postures on different backgrounds. Unlike recent systems such as SIMPLIcity [81], our prototype works entirely in a very high dimensional feature space and does not have a semantical layer to assist searches. Although more work is needed to improve our learning algorithms, this prototype demonstrates a potential to facilitate information retrieval in three respects:

- *Eliminating the seeding requirement*. One is not required to initiate a query with "good" seed images.
- *Supporting personalized search*. One can search for a general flower concept, or a specific kind of flower. One can search for a general animal concept, or a specific kind of animal (e.g., tigers).
- *Accomplishing the above tasks quickly and accurately*.

2 Relationship of Proposed Research to Current Knowledge

As discussed in Section 1, the traditional machine-learning and relevance-feedback methods are not suitable for online concept learning because of time, sample, and seeding constraints.

2.1 Related Work in Machine Learning

Most machine learning tasks can be performed offline. Therefore, time is available to sample more representative training data for learning more general rules. However, in online applications for which impatient users cannot be expected to provide a great deal of feedback, a learner must adjust to a scarcity of training data and a limited amount of training time.

Of late, ensemble techniques such as *bagging* [9], *arcing* [10], and *boosting* [70, 30] have been proposed to improve learning accuracy for decision trees and neural networks. These ensemble schemes enjoy success in improving classification accuracy through bias or variance reduction, but they do not help reduce the number of samples required to learn a query concept. In fact, most ensemble schemes actually increase learning time because they introduce learning redundancy for improving prediction accuracy [24, 30, 40, 58].

To reduce the number of required samples, the machine learning community have conducted several studies of active learning for classification. Active learning can be modeled formally as follows [13]: Given an unlabeled dataset S, an active learner L has three components: f, s, and X. The f component is a classifier that is trained on the current set of labeled data. The second component $s(X)$ is the sampling function that, given a current labeled set X, decides which subset u in U to select to query the user. The active learner returns a new f after each round of relevance feedback. The sampling techniques employed by the active learner determine the selection of the next batch of unlabeled instances to be labeled by the user.

The *query by committee* (QBC) algorithm [71, 27] is a representative active learning scheme. QBC uses a distribution over all possible classifiers and attempts greedily to reduce the entropy of this distribution. This general purpose algorithm has been applied in a number of domains (although, to our knowledge, not to the image retrieval domain) using classifiers (such as Naive Bayes classifiers [23, 55]) for which specifying and sampling classifiers from a distribution is natural. Probabilistic models such as the Naive Bayes classifier provide interpretable results and principled ways to incorporate prior knowledge. However, they typically do not perform as well as discriminative methods such as SVMs [41, 25]. (See our SVM-based approach in Section 3.1.2.)

Three major challenges of applying the QBC algorithm to a learning task are 1) Bootstrapping the algorithm by finding some initial positive examples, 2) finding useful committee members, and 3) finding informative unlabeled instances efficiently. The QBC algorithm requires at least some positive and some negative instances to initialize the learning process. However, finding positive examples that match the user's concept may be difficult, especially when concept complexity is high (i.e., when there are too many candidate concepts). (See our approach to tackle the above challenges in Section 3.1.1.)

For image retrieval, the PicHunter system [22] uses Bayesian prediction to infer the goal image, based upon the users' actions. The system shows that employing active learning can drastically cut down the number of iterations (up to 80% in some experiments). However the authors also point out that their scheme is computationally intensive, since it must recompute the conditional probability for all unlabeled samples after each round of user feedback and hence does not scale well with a sizable dataset. (See our plans to address the scalability issues in Section 3.2.)

2.2 Related Work in Information Retrieval

Relevance feedback techniques proposed by the Information Retrieval (IR) community do perform non-random sampling [43]. The study of [65] puts these query refinement approaches into three categories: *query reweighting*, *query point movement* and *query expansion*.

- *Query reweighting* and *query point movement* [39, 59, 64]. Both query reweighting and query point movement use nearest-neighbor sampling: They return top ranked objects to be examined by the user and then refine the query based on the user's feedback. If the initial query example is good, this nearest-neighbor sampling approach works fine. Unfortunately, most users do not have a good example to start a query. Refining a search around bad examples is analogous to trying to find oranges in the middle of an apple orchard by refining one's search to a few rows of apple trees at a time. It will take a long time to find oranges (the desired result).
- *Query expansion* [65, 86]. The *query expansion* approach can be regarded as a multiple-instances sampling approach. The samples of a subsequent round are selected from the neighborhood (not necessarily the nearest ones) of the positive-labeled instances of the previous round. The study of [65] shows that query expansion achieves only a slim margin of improvement (about 10% in precision/recall) over query point movement.

All traditional relevance feedback methods require seeding the methods with "good" positive examples [26, 31, 35, 54, 73, 74, 82]. For instance, sunset images must be supplied as examples in order to search for sunset pictures. As we stated in Section 1, finding good examples is the job of a search engine itself, and this circular requirement leaves the core problem—understanding users' query concepts—unsolved.

Our proposed MEGA algorithm does not require seeding a query. MEGA presents randomly selected images as the first round of examples. Even if all of the images generated in the first round are irrelevant, MEGA uses the negative images to reduce substantially the set of potentially relevant images (the the candidate sampling space can be shrunk substantially by the negative images). Thus, the probability is higher that a relevant image will be generated in the next round.

3 Plan of Work

As discussed in the previous section, the traditional machine learning algorithms and relevance feedback methods may not be adequate for supporting effective and efficient online concept learning. In this section,

Filter Name	Resolution	Representation
Color Masks	Coarse	Appearance of culture colors
Color Spread	Coarse	Spatial concentration of a color
Color Elongation	Coarse	Shape of a color
Color Histograms	Medium	Distribution of colors
Color Average	Medium	Similarity comparison within the same culture color
Color Variance	Fine	Similarity comparison within the same culture color

Table 1: Multi-resolution Color Features.

we first present our preliminary research results to set up a context for discussion. We then outline our research development plans.

3.1 Preliminary Results

We describe MEGA and SVM$_{Active}$, two online learning algorithms that we have recently developed. MEGA is an inductive-based active learning algorithm; SVM$_{Active}$ is a margin-based active learning algorithm. For detailed algorithms and extensive experimental results, please refer to [17, 77].

Table 1 and Figure 2 list the multi-resolution color and texture features that MEGA and SVM$_{Active}$ employ for learning image query-concepts. Some individual features[1] such as color histograms may be primitive. However, when organized in a hierarchical/multi-resolution manner, they can be flexibly chosen by a learning algorithm to model a subjective concept in different resolutions. The challenge for MEGA and SVM$_{Active}$ is to learn a query concept in this high-dimensional feature space both quickly and with a small number of training instances.

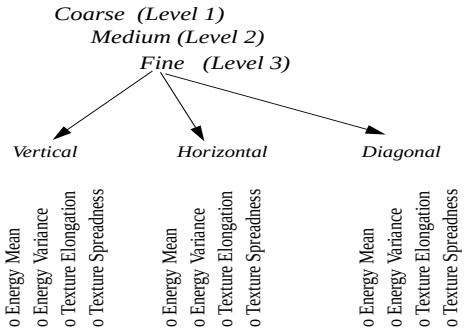


Figure 2: Multi-resolution Texture Features.

3.1.1 MEGA

The Maximizing Expected Generalization Algorithm (MEGA) models query concepts in k-CNF [45], which can formulate virtually all practical query concepts[2]. MEGA uses k-DNF to bound the sampling space from which to select the most informative samples to solicit user feedback.

[1] This project does not research feature-extraction techniques. We organize useful features (e.g., [12, 33, 34, 36, 51, 53, 62, 63, 68, 72, 76, 83]) in a hierarchical/multi-resolution manner to assist effective and efficient learning.

[2] k-CNF is more expressive than k-term DNF, and it has both polynomial sample complexity and time complexity [46, 56].

Definition 1: k-CNF: For constant k, the representation class k-CNF consists of Boolean formulae of the form $c_1 \wedge \cdots \wedge c_\theta$, where each c_i is a disjunction of at most k literals over the Boolean variables $x_1, \ldots, x_n$. (A Boolean variable represents an image feature depicted in Table 1 and Figure 2.) No prior bound is placed on θ.

Definition 2: k-DNF: For constant k, the representation class k-DNF consists of Boolean formulae of the form $d_1 \vee \cdots \vee d_\theta$, where each d_i is a conjunction of at most k literals over the Boolean variables $x_1, \ldots, x_n$. No prior bound is placed on θ.

MEGA initializes the query concept-space (QCS) as a k-CNF and the candidate concept-space (CCS) as a k-DNF. The QCS starts as the most specific concept and the CCS as the most general concept. The target concept that the learner learns is more general than the initial QCS and more specific than the initial CCS. The learner learns the QCS, while at the same time refining the CCS to delimit the boundary of the sampling space. (The shaded area in Figure 3 shows the sampling space that is between the QCS and the CCS.)

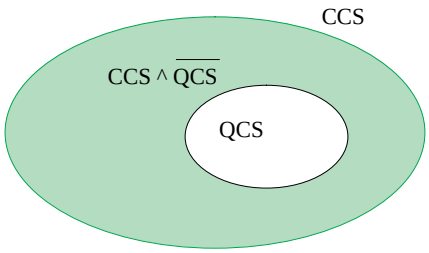


Figure 3: MEGA's Sampling Space: $CCS \wedge \overline{QCS}$.

Intuitively, at a given stage, we have a QCS and a CCS, representing the boundary of the candidate concept-space. To make sure that an example is most useful, we have tested two strategies.

1. Bounding the sample space: Avoid choosing useless unlabeled instances by using the CCS and QCS to delimit the sampling boundary.

2. Maximizing the usefulness of a sample: Choose an example that shall remove the maximum expected number of disjunctive terms from QCS. In other words, we choose an example that can maximize the expected generalization of the concept. If the example is labeled negative by the user, it can be useful to remove conjunctive terms in the CCS.

It may appear that if we pick an example that has more dissimilar disjunctions (compared to the QCS), we may have a better chance of eliminating more disjunctive terms. This is, however, not true. An example must be labeled by the user as positive to be useful to refine QCS. Unfortunately, an example that has more disjunctions that are dissimilar to the target concept is less likely to be labeled positive. Therefore, there is a tradeoff between choosing an example that has more contradictory terms and choosing an example that is more likely to be labeled positive.

Let Ψ denote the number of disjunctions remaining in the concept. The number of disjunctions that can be eliminated in the current round of sampling, denoted as ψ, is between zero and Ψ. We can write the probability of eliminating ψ terms as $P_e(\psi)$. $P_e(\psi)$ is a monotonically decreasing function of ψ. We attempt to find the ψ that can eliminate the maximum expected number of disjunctive terms given an example. Our objective function can be written as

$$\psi^* = arg_\psi \max E(\psi) = \psi \times P_e(\psi). \tag{1}$$

To solve ψ^*, we must know $P_e(\psi)$, which can be estimated by the two methods below: probabilistic estimation and empirical estimation. (Please refer to [17] for detailed discussion.) We have compared MEGA with the five traditional sampling schemes: *random*, *bounded random*, *nearest neighbor*, *query expansion*

and *aggressive*. Our preliminary experimental results show that MEGA substantially outperforms traditional sampling schemes in all query scenarios when the number of labeled instances is limited. Our prototype [15] employing MEGA can grasp a query concept in three to five rounds of feedback in a 200-feature dimensional space.

3.1.2 SVM$_{Active}$

SVM$_{Active}$ is a method that we have developed for supporting online concept learning [77]. SVM$_{Active}$ combines active learning with support vector machines (SVMs). SVMs have strong theoretical foundations and excellent empirical successes. They have been applied to tasks such as handwritten digit recognition [80], object recognition [61], and text classification [41].

For our the purpose of query-concept learning, we consider SVMs in the binary classification setting. We are given training data $\{\mathbf{x_1} \ldots \mathbf{x_n}\}$ that are vectors in some space $\mathcal{X} \subseteq \mathbb{R}^{\ell}$. We are also given their labels $\{y_1 \ldots y_n\}$ where $y_i \in \{-1, 1\}$. In their simplest form, SVMs are hyperplanes that separate the training data by a maximal margin. All vectors lying on one side of the hyperplane are labeled as -1, and all vectors lying on the other side are labeled as 1. The training instances that lie closest to the hyperplane are called *support vectors*. More generally, SVMs allow one to project the original training data in space $\mathcal{X}$ to a higher dimensional feature space $\mathcal{F}$ via a Mercer kernel operator K. In other words, we consider the set of classifiers of the form: $f(\mathbf{x}) = \sum_{\mathbf{i=1}}^{\mathbf{n}} \alpha_\mathbf{i} \mathbf{K}(\mathbf{x_i}, \mathbf{x})$. When $f(\mathbf{x}) \geq \mathbf{0}$ we classify $\mathbf{x}$ as $+1$, otherwise we classify $\mathbf{x}$ as -1.

When K satisfies Mercer's condition [11] we can write: $K(\mathbf{u}, \mathbf{v}) = \mathbf{\Phi(u)} \cdot \mathbf{\Phi(v)}$ where $\Phi : \mathcal{X} \rightarrow \mathcal{F}$ and "$\cdot$" denotes an inner product. We can then rewrite f as:

$$f(\mathbf{x}) = \mathbf{w} \cdot \mathbf{\Phi(x)}, \text{ where } \mathbf{w} = \sum_{\mathbf{i=1}}^{\mathbf{n}} \alpha_\mathbf{i} \mathbf{\Phi(x_i)}. \tag{2}$$

Thus, by using K we are implicitly projecting the training data into a different (often higher dimensional) feature space $\mathcal{F}$. The SVM then computes the α_is that correspond to the maximal margin hyperplane in $\mathcal{F}$. By choosing different kernel functions we can implicitly project the training data from $\mathcal{X}$ into spaces $\mathcal{F}$ for which hyperplanes in $\mathcal{F}$ correspond to more complex decision boundaries in the original space $\mathcal{X}$.

Intuitively, SVM$_{Active}$ works by combining the following three ideas:

1. SVM$_{Active}$ regards the task of learning a target concept as one of learning an SVM binary classifier. An SVM captures the query concept by separating the relevant images from the irrelevant images with a hyperplane in a projected space, usually a very high-dimensional one. The projected points on one side of the hyperplane are considered relevant to the query concept and the rest irrelevant.

2. SVM$_{Active}$ learns the classifier quickly via active learning. The active part of SVM$_{Active}$ selects the most informative instances with which to train the SVM classifier. This step ensures fast convergence to the query concept in a small number of feedback rounds.

3. Once the classifier is trained, SVM$_{Active}$ returns the top-k most relevant images. These are the k images farthest from the hyperplane on the query concept side.

Let V denote the uncertain concept space. The current version of SVM$_{Active}$ employs Tong & Koller's lemma [78], which chooses a pool-query that halves the uncertain space. In reality, however, finding the pool-query that can divide the uncertain space in half can be difficult and computationally intensive. Given an unlabeled instance $\mathbf{x}$ from the pool, it is not practical to explicitly compute the sizes of the new uncertain spaces $\mathcal{V}^-$ and $\mathcal{V}^+$ (i.e., the uncertain spaces obtained when $\mathbf{x}$ is labeled as -1 and $+1$ respectively). We

have tried a simple method that works as follows: Learn an SVM on the existing labeled data, and choose as the next batch of samples, those closest to the hyperplane in feature space $\mathcal{F}$. Although this simple method works efficiently, it has a couple of shortcomings. We discuss these shortcomings and how we plan to address them in Section 3.2.

Once we have performed a number of rounds of querying, we will have amassed a small set of labeled instances. We learn a final SVM using these labeled instances and use this SVM as our final classifier. In the retrieval task, we wish to return instances in order of decreasing relevance. This is easy to achieve with an SVM. Recall that an SVM is a hyperplane in the feature space such that any feature vector lying on one side of the hyperplane is labeled as "relevant" and any feature vector lying on the opposite side of the hyperplane is labeled as "not relevant." We have greatest confidence in those labels when the vectors are furthest away from the hyperplane. Hence, we can rank all of the instances in the database in order of relevance, simply by computing their signed distance from the hyperplane.

3.2 Future Work

Our future work can be divided into two thrusts consisting of six initial tasks:

1. *Accuracy improvement thrust*. We will improve the effectiveness of MEGA and SVM_{Active}, and possibly lead to the development of new online learning algorithms. We plan to investigate algorithms to perform *adaptive sampling*, to explore *co-training* using MEGA and SVM_{Active}, and to detect and deal with *concept drift*.

2. *Efficiency improvement thrust*. We will ensure that the learning algorithms are scalable in feature dimension, dataset size, and concept complexity. In this regard, we plan to investigate *multiresolution learning* (for feature-dimension scalability), *image classification* (for concept-complexity scalability), and *high-dimensional indexing* (for dataset-size scalability) methods.

Task A: Adaptive Sampling

Both MEGA and SVM_{Active} rely on effective sampling to select the most informative training data to query endusers. The sampling methods of these schemes can be improved in the following two directions:

First, both schemes use the same sampling method from one user iteration to another. Our new hypothesis is that in different stages of the learning process, the learner may want to place more weight on *exploring* or on *exploiting*. Intuitively, when the learner is used to explore the uncertainty space, a more aggressive sampling approach should be used. When the learner needs to perform fine-grained concept refinement, it should employ a more conservative sampling scheme such as the traditional "nearest neighbor." One possible criterion for determining whether to employ a more aggressive or more conservative sampling scheme can be the number of instances in the positive-labeled pool. When few samples have been labeled positive, we would want the learner to be more explorative in the uncertain feature space. For example, if a red flower is the only positive-labeled instance, this does not necessarily mean that the user's query concept is red flowers. It is possible that the query concept is "something red" or "flowers of other kinds." In this case, the sampling scheme should be more explorative. Conversely, if the learner has collected a fairly large number of positive-labeled instances that are similar to each other, then the preferred sampling scheme would be fine grained refinement. We will first conduct extensive experiments to examine whether the above hypothesis holds. If it does hold, we will develop a theoretical framework for adaptively determining the appropriate sampling scheme.

The second research direction is to design more effective sampling schemes. Once a user determines that the learner should be more explorative, it needs a robust sampling method to find the most appropriate samples to quickly reduce the uncertain space. SVM$_{Active}$ currently employs Tong & Koller's *simple method* [78], which selects all samples nearest the hyperplane that divides the positive- and negative-labeled instances. However, this theorem has been found to work optimally only when samples are taken one at a time. But as we observed in Section 3.1.2, it is necessary to present a reasonable number of samples to query users in each user iteration, since most users will not engage in even a moderate number of feedback rounds.

One of our conjectures is considering *transduction*. Transduction takes advantage of not only the labeled instances but also the unlabeled instances to learn a user's query concept. Intuitively, the unlabeled instances are likely to be clustered. This tendency provides us with some extra information which we may use to refine our idea of the user's query concept. Transduction has been successfully applied to regular passive SVMs for text classification [42] showing that using unlabeled data can indeed improve performance. An exciting direction for new research is to see whether transduction can be applied to the image retrieval domain and furthermore, whether it can also be combined with active learning. If so, we may see even greater improvements in performance for the image retrieval task.

Task B: Co-training

MEGA is an inductive-based active learning algorithm, and SVM$_{Active}$ is a margin-based one. MEGA and SVM$_{Active}$ can complement each other in at least two ways. First, SVM$_{Active}$ requires seeding at least one positive and one negative sample to initiate learning a query concept. Finding a positive sample can be very difficult when the concept complexity is high. For instance, finding a flower image in a large and diversified dataset containing thousands of image categories can take quite a while. MEGA, on the other hand, does not require seeding. If all of the images generated in the first round of feedback are found irrelevant, it will use the negative images to reduce the set of potentially relevant images substantially; thus the probability is higher that a relevant image will be generated in the next round. The logical design choice is using MEGA to bootstrap a learning session until at least one sample is labeled positive.

After identifying some positive and some negative labeled instances, SVM$_{Active}$ employs a well-founded sample selection method (at least so far) for selecting data objects nearest the hyperplane that divides positive from negative instances. MEGA, on the other hand, depends more on heuristics in choosing samples that can eliminate the maximum expected number of disjunctive terms. It is logical for two schemes to exchange sampling information, and if these methods can be adapted to work in a complementary manner, even more effective learning may result.

Task C: Concept-drift Detection and Handling

A user may alter his/her query concept in the middle of a search. Such a concept drift might be a complete change of mind or a refinement of the original target concept. For instance, a person searching for landscape images may be attracted by animal images and decide to diverge from his/her initial search path. Or one who begins with architecture images may decide to zoom into ancient cathedrals. An effective learner must be able to detect and deal with concept drift.

Concept drift can be measured by overlapping of concepts. Allen [2] proposes measuring concept overlapping by the percentage of common documents produced in the results of two consecutive user iterations. In addition, we believe that concept drift can be measured by 1) directly comparing two consecutive concepts, and 2) examining "relevant" instances. For SVM$_{Active}$, our conjecture is that the change of the norm of the hyperplane from one user iteration to the next could indicate the degree of concept drift. For MEGA, detecting

concept drift by analyzing the relevant instances may be more appropriate, since MEGA is an inductive-based algorithm. Of course, the margin-based approach of SVM*Active* and the inductive-based approach of MEGA may complement each other to produce a better detection scheme.

Task D: Multi-resolution Learning

MEGA can suffer from computational complexity when the feature dimension is high. The number of disjunctions in a k-CNF (and, likewise, the conjunctives in a k-DNF) can be written as $\sum_{i=1}^{k} \binom{M}{i}$. When M is large, a moderate k can result in a large number of disjunctive terms in a k-CNF, which burdens the learner with high space and time complexity. To reduce the number of terms in a k-CNF, we plan to divide a learning task into G sub-tasks, each of which learns a subset of the features. Dividing a feature space into G subspaces reduces both space and time complexity by a factor of $O(G^{k-1})$. The savings, in both memory space and learning time, can be considerable.

In some cases, this divide-and-conquer approach may trade precision for speed, since some terms that involve features from more than one feature subset can no longer be included in a concept. For instance, if we divide features a, b, c, and d into two groups $\{a, b\}$ and $\{c, d\}$, a 2-CNF can no longer have the following four terms: $a \vee c$, $a \vee d$, $b \vee c$, and $b \vee d$. Missing terms may or may not hurt search accuracy. If a query concept does not contain the missing terms, its search accuracy is not affected; conversely, its accuracy decreases. If we know a term is not relevant to any search concept, we can comfortably separate its variables into different groups. We plan to tackle this problem in two directions.

First, we plan to apply genetic algorithms (GAs) [29] to search for the best feature grouping. Second, we believe that an effective way to deal with the curse of dimensionality is to take advantage of the multi-resolution feature structure[3]. A learning task can start from coarse features. Once a number of coarse features have been eliminated, the learner can replace the remaining features with finer features. The positive-labeled instances used in eliminating some coarse features can be reused to eliminate some finer features.

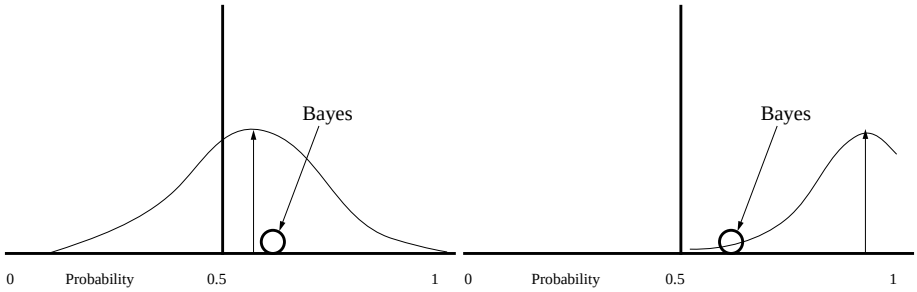
Task E: Image Classification

We have so far assumed that the learner has no prior knowledge of the dataset. It may be helpful to pre-classify the dataset into semantical categories, so the sampling scheme can take advantage of the semantical information [81]. For instance, if images are pre-labeled within their categories, and similar samples belonging to the same category have been labeled positive, the learner can conclude that the category matches the query concept.

Although pre-classifying data may offer some advantages, it also presents potential pitfalls. If the user's query concept does not map well with the pre-determined semantical structure organized by the search engine, pre-classification can be counter-productive. For example, most users may not always agree with Yahoo's ontologies (i.e., taxonomic hierarchies of classes). Not even search engines share the same ontologies among themselves. We will investigate how semantical information can be used productively to enhance an image search.

Another possible pitfall of pre-classification is that inaccurate classification can lead to poor search accuracy. We thus plan to investigate effective K-class image classification schemes. The regression approach considers the discrimination problem of K classes simultaneously; but it can be computationally intensive ($O(K^3)$) and may impose a universal assumption on all class boundaries (see Friedman [28]). Of

[3] Both text-document and multimedia features have multi-resolution characteristics. For instance, the word "landscape" is more general than the word "lake" and "color set" yields a coarser image than "color histograms."



(a) Weak Classifier (b) Strong Classifier

Figure 4: Variance vs. Margins

late, many studies have proposed ensemble schemes, which use binary (two-class) classification algorithms to solve K-class classification problems. Decomposing a K-class classification problem into a number of binary classification problems allows an ensemble scheme to model binary class boundaries with much greater flexibility at a lower computational cost (in the order of $O(K^2)$) than the regression approach. We plan to study how the SVM-based binary classifiers can be effectively combined to tackle the multi-class image classification problem.

More specifically, our initial plan is to to make improvements on several ensemble schemes, including OPC (one per class), PWC (pairwise coupling) [58], and ECOC (error-correction output coding) [24, 79]. OPC is also known as "one against others." OPC trains K binary classifiers, each of which separates one class from the other $(K-1)$ classes. PWC constructs $K(K-1)/2$ pairwise binary classifiers. The classifying decision is made by aggregating the outputs of the pairwise classifiers. ECOC was proposed by Dietterich and Bakiri [24] to reduce classification error by exploiting the redundancy of the coding scheme. Our preliminary study [?] provides us good insights into these schemes. To enhance these ensemble schemes' accuracy, we make two conjectures. First, we plan to improve the margins (i.e., confidence defined by Schapire [70]) of the SVM-based binary classifiers. Improving a classifier's confidence can reduce the classification error caused by variance. To illustrate, Figure 4 shows two classifiers. The x-axis in the figure shows the probability that point X is predicted to be in class ω_i. As long as the probability is greater than 0.5, point X is said to be in class ω_i. The circle in each subfigure indicates the prediction of Bayes. In this example, the predictions of both classifiers are consistent with Bayes and hence accurate. However, the weak prediction in Figure 4(a) has a high probability of being swayed by variance to the other side of 0.5 (resulting in a wrong prediction). Conversely, the confident prediction of the classifier in Figure 4(b) is unlikely to be affected by variance.

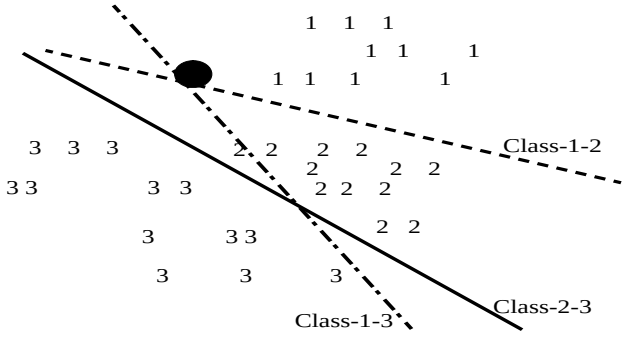


Figure 5: A Noisy Irrelevant Classifier Example.

Second, we will devise methods for removing the noise of irrelevant classifiers from class prediction. Figure 5 shows why an irrelevant classifier may display high confidence and cause misclassification. The shaded test point sits on two class boundaries: class $1-2$ and class $1-3$. Both class $1-2$ and class $1-3$ classifiers weakly predict that the point belongs to class 1. However, the class $2-3$ classifier in this example can confidently say that the point is in class 2. This is a classification variance problem to which the PWC ensemble scheme is particularly vulnerable. We will find solutions to remedy this problem. (We will also employ traditional variance reduction techniques such as *bagging* [9], *arcing* [10], and *boosting* [70, 30] in our classifiers to improve class-prediction accuracy.)

Task F: High-dimensional Indexing

The running time of MEGA and SVM$_{Active}$ scales linearly with the size of the image database both for the relevance feedback phase and for the retrieval of top-k images. This is because, for each querying round, we have to scan through the entire database to identify the most informative samples, and in the retrieval phase we again have to scan the entire database for the top-k images most relevant to the learned concept.

Retrieving desired samples and images is achieved through similarity search. Given a lookup criterion, the search engine returns objects that are most similar to that criterion. The degree of similarity between an object and the search criterion is measured by the distance between their feature vectors. The search is performed by returning the objects that are *nearest* to the query object in high-dimensional spaces [18, 67, 75]. Nearest neighbor search in a high-dimensional space is inherently expensive for two reasons: 1) the number of IOs it takes to retrieve nearest-neighbor candidates grows exponentially with the number of dimensions, and 2) the IOs must be random and hence the disk latency is very high. These problems make the traditional tree-like structures (e.g., R*-tree [4, 32], SS-tree [85], SR-tree [44], TV-tree [52], X-tree [6], M-tree [20], and K-D-B-tree [66]) inappropriate for indexing high-dimensional data. Some schemes recently developed [5, 8, 19, 60] show improvement over the tree-like structures. However, theoretical studies [3, 37, 47, 84] have confirmed the cost of exact search, independent of the data structure used. In particular, it has been shown that if N is the size of a dataset, D is the dimension, and $D >> log\ N$, then no nearest neighbor algorithm can be significantly faster than a linear search. This is known as "the curse of dimensionality."

Because of the curse of dimensionality, many recent studies suggest doing only ε-approximate nearest neighbor searches, for $\varepsilon > 0$ [3, 21, 37, 47]. Let d denote the function computing the distance between two points. We say that $p \in P$ is an ε-approximate nearest neighbor of q if for all $p' \in P$ we have $d(p,q) \leq (1+\varepsilon)d(p',q)$. Indyk and Motwani [37, 38] propose ε-approximate algorithms that enjoy improved performance bounds compared to algorithms developed by Kleinberg [47]. But the study of Kushilevitz et al. [48] points out that the recall of the algorithms proposed in [38] can be low since the search distance can be increased from a very small δ to something that is much larger than $(1+\varepsilon)\delta$.

Retrieval speed of samples and most relevant images in large databases can be improved significantly by using intelligent clustering and indexing schemes. These schemes first group similar objects into clusters. To improve IO efficiency, each cluster is then stored in a sequential file, and a mapping table is built for indexing the clusters. To answer a query, clusters near the query point are retrieved into main memory. The search engine then ranks objects in the retrieved clusters according to their distances from the query object, and returns the closest (top-k) objects as the result. Our extensive study [50] shows that the effectiveness of this clustering-for-indexing approach depends on the ability to find the clusters that contain the top k objects. Traditional cluster lookup is performed by comparing the distance between the query object and the cluster centroids. However, since a cluster may not be in a "roundish" shape, using centroids may not be reliable. We plan to use SVMs to find nearest clusters.

We plan first to group data objects into clusters. Next, we will use the SVM-OPC classifier described

in the previous section to find support vectors between clusters. We can then treat the problem of finding the nearest clusters as a classification problem: We use the SVM-OPC classifier to identify the most likely clusters to which the query object belongs. We then find the top-k results in these clusters. Of course, we will not discard the centroid information. We will experiment with different methods considering both centroids and support vectors for effectively locating clusters that contain the top-k results.

Other Tasks

The above tasks are only the beginning. Since these tasks are inter-dependent, we believe that new challenges and opportunities will arise. We are confident that we will fulfill our commitment, and the same time, identify additional open issues for later stages of the project and develop plans to tackle them. In addition to the above tasks, we plan also to 1) integrate text attributes in image searching, 2) apply the developed technologies to other data types, such as music and video, and 3) complete a fully integrated prototype that includes all proposed features.

3.3 Schedule, Partners, and Evaluation Metrics

Five of my seven Ph.D. students (currently supported by our partners) have committed to working on this project. Beitao Li and Huaxin You started on this project in Fall, 1999. They have accumulated substantial knowledge and experience for conducting advanced research, specifically in concept drift, multi-resolution learning, and image classification. Three recently joined Ph.D. students, Gang Wu, Yi Wu, and Meng Yan will focus initially on the evaluation and performance enhancement aspects of the project.

Table 2 lists the tasks and their tentative schedule for the first three years. These work projects should lead to the completion of two Ph.D. dissertations. Due to space limitations, the description below is brief. Also, note that for each task, there are others, not shown below, for the associated experimental evaluations, report generation, and technology transfer.

Task	*Year 1*	*Year 2*	*Year 3*
A. Adaptive Sampling	*Tune* MEGA & SVM_{Active} *sampling schemes*	*Devise new margin-based sampling schemes*	
B. Co-training	*Seed* SVM_{Active} *with* MEGA	*Explore* MEGA & SVM_{Active} *co-training*	*Investigate other co-training models*
C. Concept Drift Detection	*Develop concept-drift models*	*Handle* MEGA *concept drift*	*Handle* SVM_{Active} *concept drift*
D. Multi-resolution Learning	*Design* MEGA *multi-resolution learning*	*Design* SVM_{Active} *multi-resolution learning*	*Investigate video features*
E. Image Classification	*Evaluate ensemble models*	*Design new margin-based ensemble schemes*	*Integrate text into image classification*
F. High-dimensional Indexing	*Use SVMs to compute cluster boundaries*	*Evaluate centroid and margin integrated schemes*	*Support adaptive indexing*
Evaluation	*Set up testbeds*	*Integrate testbeds with partners*	*Standardize testbeds*
Prototyping	*Complete Image System Version 1*	*Complete Image System Version 2*	*Complete Video System Version 1*

Table 2: Tasks and Milestones of the First Three Years.

Table 3 lists the partners that will collaborate with us on this project. (Please see attached letters.) These

partners will collaborate in the areas of expertise, funding and evaluation.

Institution	*Research Interests*	*Main Contribution*
California DiMI Program	*multimedia project funding*	*matching fund*
IBM Almaden Research Center	*multimedia storage*	*usability study and fellowship*
IBM T.J. Watson Research Center	*image system research*	*testbeds and cash*
Sony Electronics	*multimedia content producer/provider*	*users and cash*

Table 3: Partners, Interests, and Contributions.

Evaluation Metrics

Initially, the technologies that emerge from this project will be evaluated using traditional performance, information retrieval and ease-of-use metrics. In addition to using the traditional precision and recall to measure the learning performance, we plan to employ other metrics (e.g., *quality of information* proposed in [87]) to evaluate our algorithms. We will also work with our partners, IBM research centers and Benchathlon, to identify, set up , and use standard testbeds for objective evaluation.

4 Educational Plan

My teaching goal at the University of California at Santa Barbara is to expand and modernize the undergraduate and graduate curriculum in multimedia computing and information technology. Since I joined U.C. Santa Barbara in 1999, I have designed and taught three new courses: *Hardware Software Interface, Internet Computing and Web Technology*, and *Multimedia Computing*.

4.1 Methods

My goal is to teach students about the known and inspire them to explore the unknown so that they learn not only how to build and use tools for solving real-world problems but also how to think critically and strategically about the productive processes in which they are involved. I hope to achieve this objective by 1) coupling teaching and research and 2) designing a multi-disciplinary curriculum.

- Coupling teaching and research: I use textbooks to cover the more mature topics and teach the newer topics using recent research papers. I sponsor course projects that involve development and evaluation aspects of my research projects. These methods allow students to apply what they learn to solving real-world problems, and to identify interesting open areas that might lead to new research activities.
- Designing a multidisciplinary curriculum: Extending the boundary of a well-defined field often brings to the field new research paradigms and methods. I invite guest lecturers from other departments and introduce new perspectives for viewing a problem by explaining how other fields view or solve similar problems.

In what follows I present two specific examples to demonstrate how I have employed these methods in designing and teaching new courses.

Example 1: Coupling Education and Research

Last winter, I conducted a distributed lecture course (i.e., a course taught simultaneously at two sites) with professor Patrick Hanrahan at Stanford University. We taught the “Internet Computing and Web Technology”

course via an Internet-2 link in order to share lectures and projects. The iRoom at Stanford and the ECE conference room at UCSB were connected with a 20 Mbps link. The instructor video and course slides were projected to the front of the room, and the remote audience to the back. The faculty and students at both sites could interact in real-time.

Professor Hanrahan's lectures covered the technology for producing a lecture (e.g., camera tracking, camera control, video panoramas, etc.). My lectures covered the technology for playing back lectures (e.g., interactive DTV standards and digital VCR). The course was designed in such a way that the students were taught about the most recent research results and initiatives in Internet Computing, and at the same time they were asked to advance the technology by working on projects to improve the quality of instruction. Indeed, at the end of the quarter, the results of the Stanford students' lecture-production projects and the UCSB students' lecture-playback projects enhanced the quality of the lectures from two complementary ends.

This course demonstrates a model in which students can acquire textbook knowledge, can design and experiment, and at the same time can evaluate their own work by implementing it in practice. This model tightly and effectively integrates teaching and research.

Example 2: Multidisciplinary Curriculum

The undergraduate students in my Multimedia Computing course not only learned about data compression and management techniques, but also were introduced to the psychological and artistic aspects of media data. The psychological aspect inspired new interpretation of media data. Some students worked on a project that connects human emotions with colors to depict the mood of a picture. The artistic aspect examined why an artist used, say, color and perspective in a particular way, rather than just analyzing the frequency components of a picture. Some students were inspired by Alberti's principles of perspective and devised algorithms to extract qualitative 3D information for enhancing image characterization. The results of these projects may not all be sound, but I was impressed by the level of enthusiasm that the students showed in the process of exploration.

4.2 Future Work

The three courses that I have designed and taught will continue to be my main teaching assignments at U.C. Santa Barbara over the next few years. I will update the course materials every year to include advanced developments in the fields.

In addition, I am a member of the *Media Arts and Technology* teaching program and the *Cognitive Science* research program at U.C. Santa Barbara. These are interdepartmental programs that were formed by the faculty of several departments: Music, Art, Computer Science, Education, English, Electrical Engineering, Linguistics, and Psychology. I plan to invite guest lecturers from linguistics and psychology departments to provide their perspectives on query-concept learning and multimedia searches. My aim is to encourage feedback and new ideas from faculty and students from different disciplines in order to foster interdisciplinary research.

I will continue developing small-scope undergraduate and graduate course projects related to the intelligent information-retrieval project proposed here. I envision these projects to have both research and implementation components, and I expect them to demonstrate that research can be intellectually exciting and at the same time practical in solving real-world problems. As students are exposed to cutting-edge research topics and technology, they will contribute to a larger research and implementation effort, gaining valuable experience for their future professional lives.

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